

Software Requirements Specification (SRS)

Job Portal – Resume Analysis

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of the Job Portal – Resume Analysis. This system aims to streamline the recruitment process by enabling recruiters to post jobs and job seekers to upload resumes, which are automatically analyzed to calculate a job–resume match percentage.

This SRS is intended for:

- Developers
- System architects
- Testers
- Project stakeholders

1.2 Scope

The Job–Resume Matching System is a web-based application that:

- Allows users to register as **Recruiters** or **Job Seekers**
- Enables recruiters to post and manage job listings
- Allows job seekers to upload resumes
- Uses automated resume analysis to extract skills
- Calculates a match percentage between job requirements and resumes

The system focuses on improving hiring efficiency and reducing manual resume screening.

2. System Overview

The system follows a **client-server architecture** with a frontend interface for users and backend services for authentication, job management, resume processing, and matching logic.

The system supports two roles:

- **Recruiter** – Job posting and applicant review
- **Job Seeker** – Job search and resume upload

3. System Architecture

- **Frontend:** Web-based UI for user interaction
- **Backend:** RESTful APIs for authentication, job management, and resume processing
- **Database:** Stores user profiles, jobs, resumes, and match scores
- **Processing Engine:** Handles resume parsing and matching algorithms

3.1 FRONTEND MODULES

Module 1: User Authentication & Dashboard

- Login / Register
- Role-based dashboard (Admin / Recruiter / Job Seeker)

Module 2: Job Management Interface

- Recruiter job posting
- Job listing & search
- Apply button for job seekers

Module 3: Resume Upload & Match Visualization

- Resume upload UI
- Match percentage display
- Skill highlights

3.2 BACKEND MODULES (5)

Module 4: Authentication & Authorization Service

- JWT-based login
- Role validation

Module 5: Job Management Service

- Create / update / delete jobs
- Job search APIs

Module 6: Resume Management Service

- Resume upload
- Resume storage
- File validation

Module 7: Resume Analyzer Engine

- Resume parsing
- Skill extraction
- Keyword normalization

Module 8: Job–Resume Matching Engine

- Match score calculation
- Ranking logic
- Filtering based on threshold

4. Functional Requirements

4.1 User Management

- Users can register as **Job Seeker** or **Recruiter**
- Users can log in using secure authentication
- Role-based access control is enforced

4.2 Job Management

- Recruiters can create, update, and delete job postings
- Job seekers can view and search job listings
- Job seekers can apply for jobs

4.3 Resume Management

- Job seekers can upload resumes
- Supported resume formats are validated
- Uploaded resumes are securely stored

4.4 Resume Analysis

- The system parses uploaded resumes
- Extracts relevant skills and keywords
- Normalizes extracted data for consistency

4.5 Job–Resume Matching

- The system calculates a match score
- Matching is based on skills and job requirements
- Results are ranked and filtered based on thresholds

5. Non-Functional Requirements

5.1 Security

- JWT-based authentication
- Role-based authorization
- Secure storage of user data

5.2 Performance

- Resume analysis should complete within acceptable response time
- Job search results should load efficiently

5.3 Scalability

- System should handle increasing users and job listings
- Modular backend services support scaling

5.4 Usability

- Simple and intuitive UI
- Clear visualization of match percentage

5.5 Data Privacy

- User resumes are confidential
- Data access is restricted based on roles

6. Constraints

- Internet connectivity required
- Resume parsing accuracy depends on document format
- System performance depends on server resources